YINQUAN WANG

FORWARD DEPLOYED ENGINEER (FDE) | AI APPLICATION DELIVERY & MODEL DEPLOYMENT

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Ph.D. in System Science and Data Scientist specializing in the production delivery of LLM agents, predictive models, and optimization systems. Experienced across business scoping, heterogeneous data and API integration, model serving, Docker deployment, safety controls, monitoring, and iteration after go-live. Delivered projects in district heating, mine safety, transportation, and enterprise analytics, including systems supporting **130+ heat-exchange stations, 8,000+ industrial data points, and 100+ internal users**.

CORE SKILLS

Agent systems: Agent/Harness, CodeAct, ReAct, multi-agent workflows, RAG, GraphRAG

Data & ML: feature engineering, LightGBM, XGBoost, time-series forecasting, model evaluation

Decision intelligence: operations research, scheduling, policy generation, constraint modeling

Deployment: model serving, Docker, FastAPI, API integration, versioning, runtime monitoring

Engineering & safety: Python, SQL, WebSocket, OPC UA, code sandboxing, approval and validation

RL: DQN, PPO, Actor-Critic, multi-agent RL, simulation evaluation

EXPERIENCE

China Unicom Beijing Artificial Intelligence Technology Center — Aug 2025 - Present

Data Scientist | Beijing, China

- Develop agent systems, LLM applications, predictive models, and RL solutions; contribute to customer requirement alignment, systems integration, production deployment, and post-launch evaluation.

China Unicom Digital Intelligence Business Unit — Jul 2024 - Aug 2025

Data Scientist | Beijing, China

- Delivered industrial-internet, time-series forecasting, and agent projects from algorithm prototypes and model services to production applications.

SELECTED PROJECTS

Enterprise Data Science Agent Harness Platform — Core Developer | Jun 2025 - Present

- **Business delivery:** Built an agent platform for natural-language data exploration, file analysis, code execution, and report generation, supporting **100+ team members** in daily analytics and project delivery.
- **Architecture:** Implemented the agent loop, project state, context assembly, tool protocol, error recovery, and Planner/Executor/Searcher workflows; connected agents to executable Python/SQL, retrieval, files, and report tools.
- **Reliable operation:** Delivered Docker-based code sandboxing, path and AST validation, dangerous-operation interception, resource limits, approval gates, interruption recovery, and WebSocket progress updates.

Natural-Language Data Query Agent | Agentic RAG + Text-to-SQL — Core Developer | Oct 2025 - Present

- **Retrieval and reasoning:** Built query understanding and Schema Graph/GraphRAG with vector, BM25, metric, field, and similar-SQL retrieval to improve schema linking, metric matching, and join-path reasoning.
- **Model deployment:** Deployed a vLLM inference service on Ascend hardware and integrated model serving with retrieval and SQL-generation workflows.
- **Production controls:** Implemented SQL generation, syntax and safety validation, execution feedback, automatic repair, read-only enforcement, sensitive-field blocking, and low-confidence refusal.

District-Heating Predictive Control Platform — Core Developer | May 2025 - Present

- **Data and models:** Integrated OPC UA telemetry, room-temperature collection, and weather APIs across **8,000+ industrial data points**; built LightGBM models for supply temperature and pump frequency.
- **Deployment:** Integrated predictive models and control policies into production workflows with versioning, monitoring, continual training, hot updates, and Dockerized services.
- **Safe delivery and impact:** Built a predict-policy-execute-monitor-diagnose-human-fallback loop with heat-balance, equipment-boundary, and approval constraints; deployed across **130+ stations and 200+ loops**, reaching R^2 **0.98** and room-temperature deviation within **±0.5°C**.

Mine Gas-Anomaly Early-Warning System — Core R&D | Feb 2025 - Oct 2025

- **Modeling:** Built a multi-path ResNet18 fusion model using dual-channel sensor statistics, recurrence plots, and time-frequency features to detect six classes of mine-safety precursors.
- **Deployment and impact:** Packaged inference as an online warning service connected to ClickHouse streams through Flask APIs and Docker, covering **30+ key working faces**; achieved AUC **96.65%** and a **105%** recall improvement over the single-view baseline.

Domain Adaptation for Industrial-Control LLMs — Independent Research | Oct 2025 - Mar 2026

- Built 8,185 domain samples from 144 stations and three heating seasons; fine-tuned Qwen2.5-0.5B with LoRA for control-parameter parsing and few-shot adaptation.
- Reduced temperature MAE from **3.43°C to 1.10°C**, frequency MAE from **2.12 Hz to 0.51 Hz**, and increased structured parsing from **57% to 100%**.

EDUCATION

Beijing Jiaotong University | Ph.D. in System Science (combined M.S./Ph.D.) Sep 2018 - Jun 2024

- Research: ride-sourcing operations, operations research, and reinforcement learning.

Hiroshima University | Visiting Ph.D. Researcher, Transportation Planning and Management May 2022 - May 2023

- China Scholarship Council-funded joint doctoral program.

Qingdao University of Technology | B.Eng. in Traffic Engineering Sep 2014 - Jun 2018

SELECTED PUBLICATIONS

IEEE T-ITS, 2023: Y. Wang et al. **Reassignment Algorithm of the Ride-Sourcing Market Based on Reinforcement Learning.**

IEEE T-ITS, 2024: Y. Wang et al. **Promoting Collaborative Dispatching in the Ride-Sourcing Market with a Third-Party Integrator.**

Computers & Industrial Engineering, 2024: Y. Wang et al. **Reinforcement Learning-Based Order-Dispatching Optimization in Ride-Sourcing Service.**

Journal of Central South University, 2023: Y. Wang et al. **Order Dispatching Optimization in Ride-Sourcing Market by Considering Cross Service Modes.**

HONORS

National Second Prize, Digital China Innovation Competition (Team Lead, ranked 1/3), 2020

National Third Prize, Mathematical Contest in Modeling for Chinese Graduate Students, 2019

China Scholarship Council Joint-Training Doctoral Scholarship, 2021

Beijing Jiaotong University Doctoral Scholarship, 2019-2022